

SACHIN PATEL

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PROFILE SUMMARY

Operations Research Scientist with 1.5+ years building and deploying production MILP and Vehicle Routing Problem (VRP) optimization models for secondary distribution logistics at scale. Delivered end-to-end optimization pipelines in Pyomo (Gurobi, CBC, GLPK) that cut fleet operating costs and improved vehicle utilization by 10% across 150+ daily routes. Combines mathematical optimization depth with full-stack engineering - SQL/MongoDB ETL pipelines, backend route-analytics APIs, and ML-driven forecasting - plus applied LLM tooling for automated optimization model generation. Targeting Applied Scientist / Optimization Engineer roles to scale prescriptive analytics and decision-intelligence systems across supply chain and logistics.

KEY SKILLS

- **Optimization & Mathematical Programming:** MILP, Linear & Integer Programming, Vehicle Routing Problem (VRP), Network Optimization, Combinatorial Optimization, Constraint Programming, Pyomo, Gurobi, CBC, GLPK, OR-Tools, PuLP
- **Programming & Data Engineering:** Python, SQL, MongoDB, ETL Pipelines, Pandas, NumPy, Git
- **Machine Learning & AI:** Scikit-learn, Statistical Modeling, Forecasting, Large Language Models (LLMs), Prompt Engineering, Generative AI
- **Backend & Analytics:** REST APIs, Data Validation, Streamlit, Data Visualization, Excel

EXPERIENCE

Caliper Business Solutions | Operations Research Scientist **Jul 2024 - Jan 2025**

- Designed and deployed MILP and VRP optimization models in Pyomo (Gurobi in production, CBC/GLPK for prototyping) for secondary distribution logistics, reducing fleet operating costs and improving vehicle utilization by 10% across 150+ daily routes.
- Engineered backend APIs for GPS-based route analytics - including data validation and frequent-path detection across 50K+ GPS records - cutting route-analysis turnaround time by 30%.
- Built SQL and MongoDB ETL pipelines in Python (using Airflow) that ensured data integrity across optimization and analytics workflows, enabling the logistics planning team to run daily production models without data delays
- Deployed modular PuLP/OR-Tools components to production under Git version control with full test coverage; applied Scikit-learn statistical models to improve supply-chain forecasting accuracy by 8%.

Operations Research Intern - VRP Solutions **May 2023 - Jun 2024**

- Built customized VRP models in Pyomo integrating commercial (Gurobi) and open-source (CBC/GLPK) solvers, shortening route-planning cycles from 6 hrs to 2 hrs and enabling daily schedule updates.
- Automated data collection, cleaning, and preprocessing pipelines in Python; prototyped and validated routing strategies later scaled into production, contributing to a ~10% reduction in vehicle usage and operating costs.

PROJECTS

LLM-Powered Operations Research Solver | [Live Demo](#) **Mar 2026 - Apr 2026**

- Built an LLM-powered platform that translates natural-language business problems into MILP formulations, using prompt-engineered constraint/objective extraction to cut model development time from hours to minutes for non-technical users.

EV Charging Grid Orchestration | [Live Demo](#) **Jan 2026 - Mar 2026**

- Designed an EV charging infrastructure planning tool applying demand clustering and Set Cover optimization across 500+ simulated demand points, achieving 90%+ geographic coverage while minimizing station count.

Dynamic Pricing Engine (Live Demo) | [Live Demo](#) **Mar 2026 - May 2026**

- Built a pricing optimization engine combining demand forecasting, price-elasticity modeling, and mathematical optimization, delivering simulated profit improvements of up to 22% across tested scenarios.

EDUCATION

M.Sc., Industrial Engineering & Operations Research | *Indian Institute of Technology Bombay* **Jul 2022 - May 2024**

B.Sc., Operational Research | *Deen Dayal Upadhyaya College, University of Delhi* **Jul 2019 - May 2022**

ACHIEVEMENTS

- AIR 60, IIT JAM Mathematics (2022) • Elite Silver, NPTEL Data Science for Engineers • Languages: English, Hindi